

**Student Name:** Shaked Sarbstein

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**Chosen Cyber Technique:** Man In The Middle - Evil Twin

**Target Paper Citation:** L. M. da Silva, V. M. Andreghetti, R. A. F. Romero, and K. R. L. J. C. Branco, “Analysis and Identification of Evil Twin Attack through Data Science Techniques Using AWID3 Dataset,” in *Proceedings of the 6th International Conference on Machine Learning and Machine Intelligence (MLMI)*, October 2023, pp. 128–135. DOI: 10.1145/3635638.3635665.

**Paper Meta-Information & Research Group Focus**

- **Publication Venue:** MLMI 2023 – the 6th International Conference on Machine Learning and Machine Intelligence (ACM, peer-reviewed).
- **Research Group & Affiliation:** Institute of Mathematics and Computer Sciences (ICMC) and Dept. of Electrical & Computer Engineering, University of São Paulo (USP), Brazil. The group focuses on machine-learning and data-science applied to security and networked systems.
- **Funding/Sponsorship:** Partially financed by CAPES (Brazil), Finance Code 001. The work directly extends the AWID3 research of Chatzoglou et al. (the dataset's authors).

**Problem Statement & Literature Gap Analysis**

- **The Status Quo:** prior AWID2/AWID3 ML studies (Islam & Allayear, Chatzoglou et al., Sharma, Saini et al., Torres) reach 97–99% accuracy, but are predominantly multi-class and treat many attack families together.
- **The Identified Limitations:** none isolate a single attack, so it stays unclear which features discriminate the evil twin; false-positive rates stay high if untreated; severe class imbalance is under-addressed; and robustness to zero-day / adversarial variants is not tested.
- **The Core Research Question:** can the evil twin attack in AWID3 be detected at high accuracy and very low false-positive rate using a small, interpretable subset of 802.11 features instead of the full 253-feature space?

**Model Selection & Architectural Justification**

The study compares six supervised classifiers (binary: Normal vs Evil Twin), with LightGBM as the focal model:

| Selected Model / Architecture     | Core Technical Mechanism                        | Reason for Selection                                                                                |
|-----------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| LightGBM (focal model)            | Histogram-based gradient-boosted decision trees | Fast, low-memory, high accuracy on large tabular data; chosen as the model to optimize with Optuna. |
| XGBoost / Random Forest           | Boosting / bagged tree ensembles                | Strong tree baselines; RF was the best non-optimized model.                                         |
| Decision Tree / MLP / Gaussian NB | Single CART / feed-forward NN / probabilistic   | Interpretable, shallow deep-learning and simple statistical references.                             |

- **Baseline Comparisons:** beyond the six internal models, results are compared to prior AWID3 literature (Chatzoglou et al., Torres) at 97–99% accuracy.

**Hyperparameters & Model Optimization**

All six algorithms first ran with library defaults; LightGBM was then tuned with Optuna (10 trials) and validated with stratified 5-fold cross-validation. The exact tuned values (all-columns model, taken from the authors' source code) are:

| Hyperparameter        | Specific Value / Setting | Technical Justification / Method                               |
|-----------------------|--------------------------|----------------------------------------------------------------|
| lambda_l1 ; lambda_l2 | 3.53e-06 ; 0.00499       | L1/L2 regularization searched by Optuna to reduce overfitting. |
| num_leaves            | 37                       | Tree capacity – balances fit against overfitting.              |

|                                 |            |                                                           |
|---------------------------------|------------|-----------------------------------------------------------|
| min_child_samples               | 11         | Minimum samples per leaf – prevents overfit splits.       |
| feature_fraction                | 0.427      | Per-tree column subsampling – regularization and speed.   |
| bagging_fraction ; bagging_freq | 0.692 ; 11 | Row subsampling every 11 iterations – variance reduction. |

## Data Pipeline & Implementation Workflow

[Raw AWID3 CSVs] → [Filter Evil Twin + Clean/Impute] → [Encode & Scale] → [Balance] → [Train/Validate, 5-fold] → [Test / Output]

- **Data Preprocessing & Cleaning:** grouped the separate CSVs into one table; removed 25 null-label rows; imputed numerical NaNs with the mean and categorical NaNs with the most-frequent value; dropped constant and highly correlated columns (Pearson > 0.95). Min-max scaling for numerical features, One-Hot Encoding for categorical.
- **Data Balancing:** severe imbalance (3,673,854 normal vs 104,827 evil twin) handled with Random Undersampling (normal → 150% of attack count) then K-Means SMOTE, ending at 1,888 normal / 1,891 attack samples.
- **Feature Engineering / Selection:** started from the 16 expert features of Chatzoglou et al.; the top features by importance were radiotap.dbm\_antsignal, frame.len, wlan.duration (numerical) and wlan.fc.protected (categorical).
- **Software Stack:** Python, pandas, NumPy, scikit-learn, XGBoost, LightGBM, Optuna, imbalanced-learn, ydata-profiling, PyCaret.

## Experimental Environment & Quantitative Results

- **Benchmark Dataset:** AWID3 evil-twin subset (binary Normal vs Evil Twin).
- **Environment:** Intel Core i5-11300H @3.10GHz (4C/8T), 16GB RAM, 256GB SSD, Linux.
- **Evaluation Metrics:** Accuracy, Precision, Recall (TPR), F1, AUC, and False Positive Rate (FPR) from the confusion matrix.

Reported detection performance (best models):

| Model Variant                  | Accuracy | Precision | Recall (TPR) | F1     | FPR   |
|--------------------------------|----------|-----------|--------------|--------|-------|
| Proposed: LightGBM (Optimized) | 99.31%   | 98.80%    | 99.84%       | 99.32% | 1.17% |
| Best Baseline: Random Forest   | 99.34%   | 99.00%    | 99.68%       | 99.34% | 1.01% |

FPR above is on validation (same source for both models, for a fair comparison). On the held-out **test set** the paper reports LightGBM (Opt.) FPR = 0.602% with all 16 columns and 0.898% with only 4 columns – still reaching 99.29% accuracy.

## Novelty & Scientific Contribution

- **The “Secret Sauce”:** the first AWID3 study to isolate the evil twin attack and quantify which features drive its detection, showing a 4-feature interpretable model (with wlan.fc.protected as the decisive categorical signal) reaches a test FPR below 1%.
- **Theoretical vs Practical Value:** the small feature set and fast LightGBM training (~0.3s) make near-real-time / line-rate IDS use feasible, but the model is validated offline on a single dataset only – not yet deployed live.

## Actionable Engineering Takeaways for Our Project

- **Features used for detection (as seen in the logs):** antenna signal strength (radiotap.dbm\_antsignal), frame length, frame duration, and the protected-frame flag (wlan.fc.protected) – consistent with a rogue AP cloning an SSID with slightly different RF/MAC behaviour.

- **What to Adopt:** the RUS + K-Means SMOTE balancing recipe, Pearson > 0.95 redundancy pruning, and the 16→4 feature reduction as our supervised baseline (LightGBM / RF).
- **What to Avoid:** validating on a single dataset (the authors flag possible AWID3-specific overfitting) and using MLP on tabular data (slowest at ~10s with no accuracy gain).
- **Proposed Improvement:** the authors explicitly leave deep learning to future work – our planned Transformer model on the same evil-twin features, plus cross-dataset testing (e.g. AWID2), is the natural extension.